SLISOFT ACADEMY

OBJECT ORIENTED PROGRAMMING

EPISODE - ④

CONSTRUCTORS

* How does an object get initialized ?

Problem:

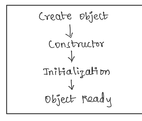
» Car 1 = Car ()

Car 1
colour = ? speed = ? fuel = ?

* We need initialization.

CONSTRUCTOR:

* A special mechanism used during object creation to initialize the object.



* Constructors syntax and rules vary by each programming language

Default Initialization:

Car () [NO arguments]

CAR OBJECT
→ speed = 0 → fuel = 0

Parameterized Constructor:

» car 1 = Car ("RED", 0, 100)

car 1
→ colour = RED → speed = 0 → fuel = 100

» car 2 = Car ("GREEN", 50, 80)

car 2
→ colour = GREEN → speed = 50 → fuel = 80

THIS

» car 1. change Speed (50)

Current object

* "this" refers to the current object

Note: Exact syntax vary by language.

For example: python uses "self" for current object

CONSTRUCTOR OVERLOADING

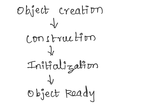
* Different ways to initialize an object

» Car ()
» Car ("green")
» Car ("Red", 0, 80)

→ Same constructor concept
→ Different parameter lists

Note: Language support varies.

RECAP



Constructor:

- ✓ Initialize object state
- ✓ Accept initial values
- ✓ Establish a valid starting state

Can anyone directly change our data ?

ENCAPSULATION

NEXT VIDEO ...